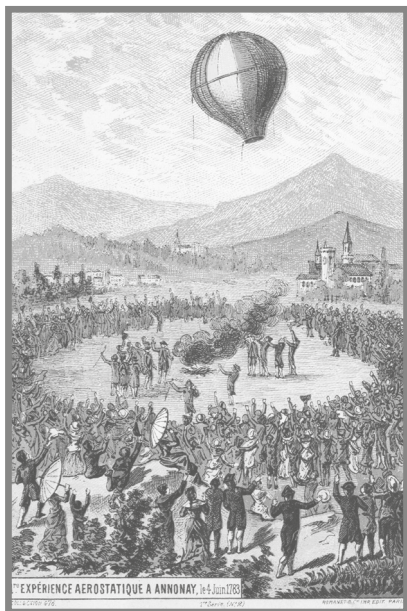


Enlightenment scientists took this investigation of reality further, making possible, for example, Swedish botanist Carl Linnaeus's development of a proper, rational biological classification in the early 18th century. The inquiring, reason-based approach of the Enlightenment also triggered dramatic technological advances. In the 1760s, the Scottish physician Joseph Black discovered carbon dioxide, while in 1769, Scotsman James Watt made improvements to the steam engine that increased its efficiency, thereby enabling the improvement of factories. The *Encyclopédie* helped to publicize these, and other, achievements of 18th-century scientists, as well as those of their precursors.

The work also found an audience in the learned societies, academies, and universities that flourished in the Enlightenment period. Although many teachers and scholars at Europe's older, Church-dominated universities remained deaf to the new scientific way of thought, more progressive ones helped teach and promote it.

Equality and freedom

The scientific revolution and the Enlightenment also encouraged the belief that reason could reveal natural laws in human affairs. Instead of drawing fact from faith, Enlightenment thinkers believed that politics should be separated from religion, that neither should curtail the rights of the individual, and that people should be free to express their opinions, worship in their own way, and read what they want to. This political doctrine, which is often labeled liberalism, had roots in the work of 17th-century philosophers such as Englishman John Locke—the father of liberalism. Locke asserted that there are certain intrinsic human rights that are not



In 1783, France's Montgolfier brothers gave the first demonstration of their new invention, the hot-air balloon, bringing science to the forefront of public attention in a spectacular way.

dependent on law or custom—in other words, they exist quite separately from what the Church or monarch might decree. These rights could be expressed in different ways, but included the right to life, the right to liberty, and the freedom to own what one has produced. These ideas were central to Enlightenment thinkers, following Locke, who felt that such natural rights should form the basis of any system of government.

Liberal ideas also found expression in the work of Enlightenment writers. For example, Voltaire, in books such as the *Philosophical Dictionary*, highlighted the injustices and abuses of the Catholic Church, and espoused values such as tolerance, freedom of the press, and the promotion of reason over doctrine and religious revelation. In his *Spirit of the Laws*, Montesquieu advocated the separation of governmental powers (legislature, executive, judiciary) and pressed for an end to slavery. In *The Social Contract*, Jean-Jacques Rousseau rejected

“In all the ages of the world, priests have been the enemies of liberty.”
David Hume